

Exercise 2: Implementing the Factory Method Pattern

Document.java:

```
public interface Document {  
    void open();  
}
```

WordDocument.java

```
public class WordDocument implements Document {  
  
    public void open() {  
        System.out.println("Word Document opened.");  
    }  
}
```

PdfDocument.java:

```
public class PdfDocument implements Document {  
  
    public void open() {  
        System.out.println("PDF Document opened.");  
    }  
}
```

ExcelDocument.java

```
public class ExcelDocument implements Document {  
  
    public void open() {  
        System.out.println("Excel Document opened.");  
    }  
}
```

DocumentFactory.java

```
public abstract class DocumentFactory {  
    public abstract Document createDocument();  
}
```

WordFactory.java

```
public class WordFactory extends DocumentFactory {  
    public Document createDocument() {  
        return new WordDocument();  
    }  
}
```

PdfFactory.java

```
public class PdfFactory extends DocumentFactory {  
    public Document createDocument() {  
        return new PdfDocument();  
    }  
}
```

ExcelFactory.java

```
public class ExcelFactory extends DocumentFactory {  
    public Document createDocument() {  
        return new ExcelDocument();  
    }  
}
```

FactoryMethodTest.java

```
public class FactoryMethodTest {  
  
    public static void main(String[] args) {  
  
        DocumentFactory wordFactory = new WordFactory();  
        Document word = wordFactory.createDocument();  
        word.open();  
  
        DocumentFactory pdfFactory = new PdfFactory();  
        Document pdf = pdfFactory.createDocument();  
        pdf.open();  
  
        DocumentFactory excelFactory = new ExcelFactory();  
        Document excel = excelFactory.createDocument();  
        excel.open();  
    }  
}
```

Output:

```
cd "/Users/adityahargunani/Documents/GitHub/Cognizant/Week 1/Design Patterns and Principles/Exercise2/FactoryMethodPatternExample/" && javac FactoryMethodTest.java && java FactoryMethodTest  
adityahargunani@Adityas-MacBook Cognizant % cd "/Users/adityahargunani/Documents/GitHub/Cognizant/Week 1/Design Patterns and Principles/Exercise2/FactoryMethodPatternExample/"  
" && javac FactoryMethodTest.java && java FactoryMethodTest  
Word Document opened.  
PDF Document opened.  
Excel Document opened.  
adityahargunani@Adityas-MacBook FactoryMethodPatternExample %
```